CI/CD Pipeline with GitHub Actions & Docker

1. Introduction

This project showcases a simple yet effective CI/CD pipeline using GitHub Actions and Docker. Whenever a developer pushes code to GitHub, the pipeline automatically builds a Docker image and uploads it to Docker Hub. This automation enhances development speed and deployment reliability.

2. Abstract

The application is a lightweight Flask web service, styled with HTML and CSS. The project uses containerization through Docker and continuous integration techniques via GitHub Actions. All components work together to achieve fast, reliable, and automated deployments.

3. Tools Used

- Python (Flask Framework)
- Docker & Docker Hub
- Git & GitHub
- GitHub Actions for CI/CD automation
- HTML/CSS for UI design

4. Steps Involved

1. Developed a basic Flask app with an HTML interface.
2. Containerized the app using Docker.
3. Pushed the app to GitHub.
4. Configured GitHub Actions to automate build and push to Docker Hub.
5. Pulled the image from Docker Hub and deployed it locally.

5.1 GitHub Actions Workflow Success

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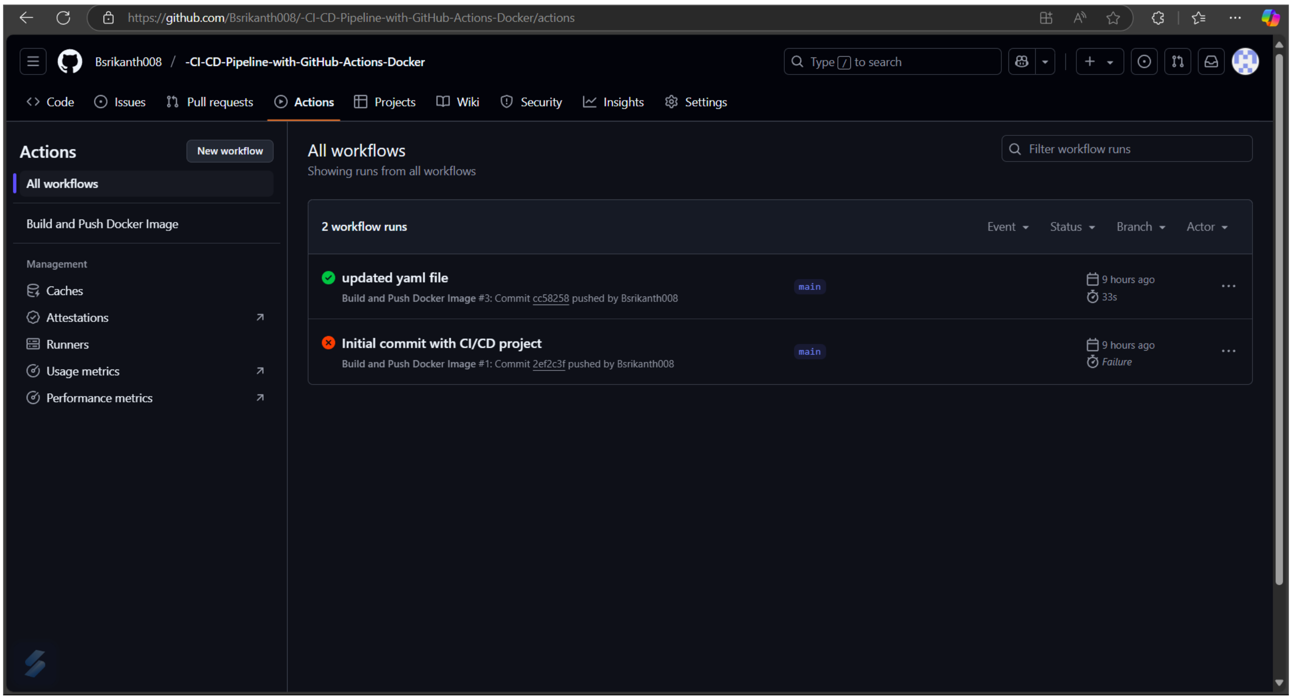
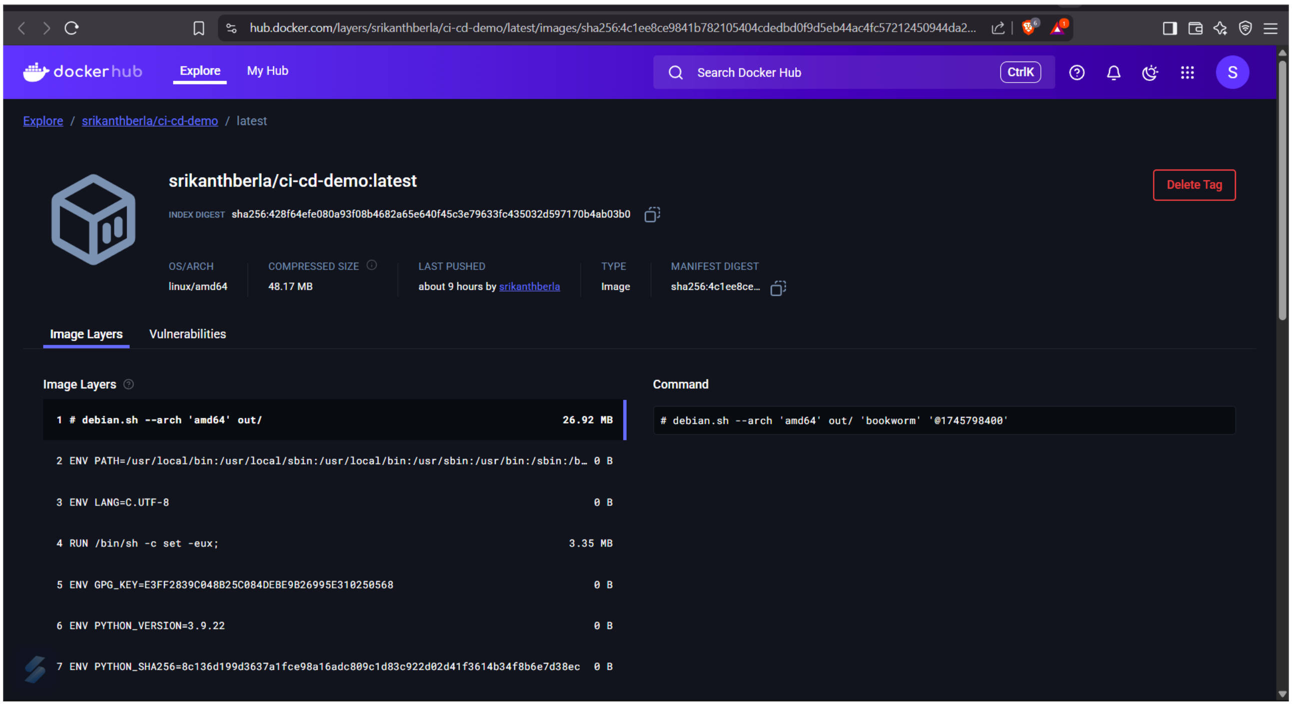


5.2 Docker Hub Repository with Image



5.3 App Running Locally

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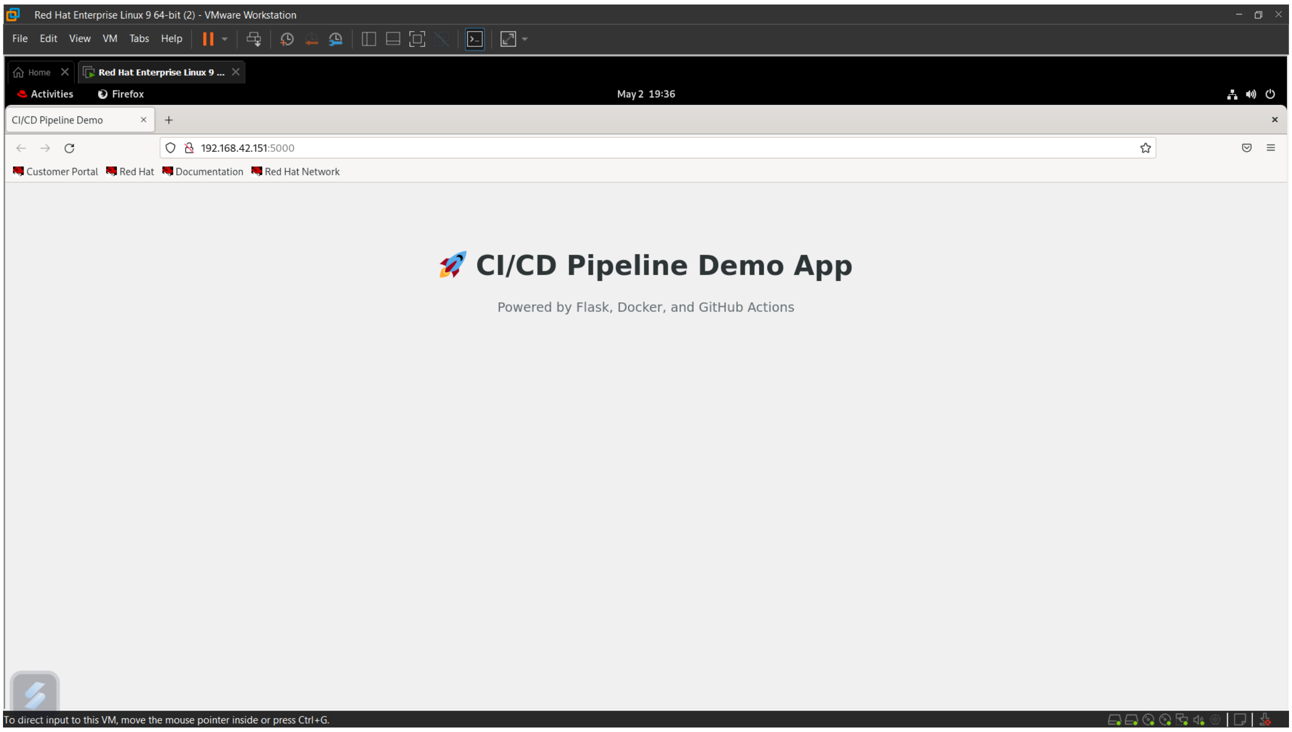

6. Conclusion

The project provided hands-on experience with modern DevOps practices, emphasizing the importance of automation in application deployment. The skills practiced here align well with real-world DevOps engineer responsibilities.